

Module 3 — Class 1 Assignment: Data Cleaning

Type-it-yourself exercise

UN AI/ML Mentorship

Module 3 — Class 1 Assignment: Data Cleaning

Topic: Cleaning — missing values, duplicates, type conversion

Goal: Open a **fresh Google Colab notebook** and type the code yourself by following the steps below. Do NOT copy-paste from the working notebook. Typing it yourself is how you learn the syntax.

How to submit

1. Open Google Colab → File → New Notebook.
 2. Type the code from each step below. Add a short # comment on EVERY cell so we know you understand it.
 3. Save the notebook as `Module<X>_Class<Y>_<YourName>_Assignment.ipynb`.
 4. Push it to your team repo at `module-<X>/class_<Y>/submissions/<YourName>/`.
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Step 0 — Load the Titanic dataset

We use a DIFFERENT dataset here (Titanic) so you practice cleaning on a new problem.

```
import pandas as pd
import numpy as np
url = 'https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv'
df = pd.read_csv(url)
print('Shape:', df.shape)
df.head()
```

Expected: Shape: (891, 12) and a 5-row preview with PassengerId, Survived, Pclass, Name, Sex, Age, etc.

Step 1 — Find missing values

What to do: Count missing values per column.

Type this in a new cell:

```
df.isnull().sum()
```

Expected output:

Age 177\nCabin 687\nEmbarked 2\n(others 0)

Step 2 — Fill missing Age with the median

What to do: Use fillna with the median of Age.

Type this in a new cell:

```
median_age = df['Age'].median()
df['Age'] = df['Age'].fillna(median_age)
print('Median used:', median_age)
print('Missing Age now:', df['Age'].isnull().sum())
```

Expected output:

Median used: 28.0\nMissing Age now: 0

Step 3 — Fill missing Embarked with the mode (most common value)

What to do: mode() returns a Series; use [0] to get the first value.

Type this in a new cell:

```
mode_embarked = df['Embarked'].mode()[0]
df['Embarked'] = df['Embarked'].fillna(mode_embarked)
print('Mode used:', mode_embarked)
print('Missing Embarked:', df['Embarked'].isnull().sum())
```

Expected output:

Mode used: S\nMissing Embarked: 0

Step 4 — Drop the Cabin column (too much missing)

What to do: Use df.drop with columns=...

Type this in a new cell:

```
df = df.drop(columns=['Cabin'])
print('Columns now:', df.shape[1])
```

Expected output:

Columns now: 11

Step 5 — Check for duplicate rows

What to do: duplicated() returns True/False per row; .sum() counts the Trues.

Type this in a new cell:

```
print('Duplicates:', df.duplicated().sum())
```

Expected output:

Duplicates: 0

Step 6 — Convert Survived from 0/1 to 'No'/'Yes'

What to do: Use `.map` with a dictionary.

Type this in a new cell:

```
df['Survived_label'] = df['Survived'].map({0: 'No', 1: 'Yes'})
df[['Survived', 'Survived_label']].head()
```

Expected output:

5 rows showing 0->No, 1->Yes.

Step 7 — Strip whitespace from Sex column

What to do: Use `str.strip()` to remove leading/trailing spaces.

Type this in a new cell:

```
df['Sex'] = df['Sex'].str.strip().str.lower()
df['Sex'].value_counts()
```

Expected output:

```
male      577\nfemale    314
```

Step 8 — Final check — confirm dataset is clean

What to do: Print `isnull` sum + shape.

Type this in a new cell:

```
print('Missing per column:')
print(df.isnull().sum())
print()
print('Final shape:', df.shape)
```

Expected output:

All 0s, Final shape: (891, 12)

Checklist before you submit

- ☐ All cells run without error
- ☐ Every code cell has a `#` comment in your own words
- ☐ Outputs are visible (you saved AFTER running)
- ☐ File name is correct: `Module<X>_Class<Y>_<YourName>_Assignment.ipynb`

Deadline: before next class.